

Solutions & Notes of Homework 5

Yang He, Junda Wu

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Tanquam ex ungue leonem.

One knows the lion by his claw.

—Johann Bernoulli

Problem 1

Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in V . Let $v \in V$. Prove that

$$\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2$$

if and only if $v \in \text{span}(e_1, \dots, e_m)$.

✓ Proof:

Since $e_1, \dots, e_m$ is an orthonormal list of vectors in V , it is linearly independent, thus it is an orthonormal basis of $\text{span}(e_1, \dots, e_m)$.

If $v \in \text{span}(e_1, \dots, e_m)$, we have $v = a_1 e_1 + \dots + a_m e_m$, where $a_1, \dots, a_m$ are scalars. Take inner product with e_k and we find that $a_k = \langle v, e_k \rangle$. Then

$$v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m.$$

Thus,

$$\begin{aligned} \|v\|^2 &= \langle \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m, \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m \rangle \\ &= \|\langle v, e_1 \rangle e_1\|^2 + \dots + \|\langle v, e_m \rangle e_m\|^2 \\ &= |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2. \end{aligned}$$

Conversely, if $\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2$, we denote $u = v - (\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m)$. Then we have

$$\langle u, e_i \rangle = \langle v, e_i \rangle - \langle v, e_i \rangle = 0, \quad i = 1, \dots, m.$$

Thus, $u \perp \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m$. Since $v = u + (\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m)$, it follows from Pythagorean theorem that

$$\|v\|^2 = \|u\|^2 + \|\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m\|^2.$$

Thus,

$$\begin{aligned}\|u\|^2 &= \|(\langle v, e_1 \rangle e_1 + \cdots + \langle v, e_m \rangle e_m)\|^2 - \|v\|^2 \\ &= \left(|\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_m \rangle|^2 \right) - \left(|\langle v, e_1 \rangle|^2 + \cdots + |\langle v, e_m \rangle|^2 \right) \\ &= 0,\end{aligned}$$

which implies $u = 0$ and further implies that

$$v = \langle v, e_1 \rangle e_1 + \cdots + \langle v, e_m \rangle e_m.$$

Hence, $v \in \text{span}(e_1, \dots, e_m)$. ■

Problem 2

Suppose n is a positive integer. Prove that

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \dots, \frac{\cos nx}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \dots, \frac{\sin nx}{\sqrt{\pi}}$$

is an orthonormal list of vectors in $C[-\pi, \pi]$, the vector space of continuous real-valued functions on $[-\pi, \pi]$ with inner product

$$\langle f, g \rangle = \int_{-\pi}^{\pi} f(x)g(x)dx.$$

✓ **Proof:** First we show that the norm of every vector in the list is 1.

$$\begin{aligned}\left\| \frac{1}{\sqrt{2\pi}} \right\| &= \left(\int_{-\pi}^{\pi} \frac{1}{2\pi} dx \right)^{1/2} = \left(\frac{1}{2\pi} \cdot 2\pi \right)^{1/2} = 1; \\ \left\| \frac{\cos kx}{\sqrt{\pi}} \right\| &= \left(\int_{-\pi}^{\pi} \frac{\cos^2 kx}{\pi} dx \right)^{1/2} = \left(\frac{1}{\pi} \int_{-\pi}^{\pi} \frac{\cos 2kx + 1}{2} dx \right)^{1/2} = 1 \quad (k \in \mathbb{N}^*). \\ \left\| \frac{\sin kx}{\sqrt{\pi}} \right\| &= \left(\int_{-\pi}^{\pi} \frac{\sin^2 kx}{\pi} dx \right)^{1/2} = \left(\frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1 - \cos 2kx}{2} dx \right)^{1/2} = 1 \quad (k \in \mathbb{N}^*). \\ &\left(\text{using } \int_{-\pi}^{\pi} \frac{\cos 2kx}{2} dx = 0 \text{ and } \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2} dx = \frac{1}{\pi} \cdot \pi = 1 \right)\end{aligned}$$

Next we show that every two different vectors in the list are orthogonal to each other.

For any pair of positive integers i, j ($i \neq j$), we have

$$\begin{aligned}\left\langle \frac{\cos ix}{\sqrt{\pi}}, \frac{\cos jx}{\sqrt{\pi}} \right\rangle &= \int_{-\pi}^{\pi} \frac{\cos ix}{\sqrt{\pi}} \cdot \frac{\cos jx}{\sqrt{\pi}} dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} \cos ix \cdot \cos jx dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} [\cos(i+j)x + \cos(i-j)x] dx \\ &= \frac{1}{2\pi} \left[\frac{\sin(i+j)x}{i+j} + \frac{\sin(i-j)x}{i-j} \right] \Bigg|_{-\pi}^{\pi} = 0.\end{aligned}$$

Similarly, $\left\langle \frac{\sin ix}{\sqrt{\pi}}, \frac{\sin jx}{\sqrt{\pi}} \right\rangle = \left\langle \frac{\cos ix}{\sqrt{\pi}}, \frac{\sin jx}{\sqrt{\pi}} \right\rangle = 0$.

For any positive integer k , we have

$$\begin{aligned} \left\langle \frac{1}{\sqrt{2\pi}}, \frac{\cos kx}{\sqrt{\pi}} \right\rangle &= \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \cdot \frac{\cos kx}{\sqrt{\pi}} dx \\ &= \frac{1}{\sqrt{2}\pi} \int_{-\pi}^{\pi} \cos kx dx \\ &= \frac{1}{\sqrt{2}\pi} \left[\frac{1}{k} \sin kx \right]_{-\pi}^{\pi} = 0, \\ \left\langle \frac{1}{\sqrt{2\pi}}, \frac{\sin kx}{\sqrt{\pi}} \right\rangle &= \int_{-\pi}^{\pi} \frac{1}{\sqrt{2\pi}} \cdot \frac{\sin kx}{\sqrt{\pi}} dx \\ &= \frac{1}{\sqrt{2}\pi} \int_{-\pi}^{\pi} \sin kx dx \\ &= \frac{1}{\sqrt{2}\pi} \left[-\frac{1}{k} \cos kx \right]_{-\pi}^{\pi} = 0, \end{aligned}$$

and

$$\begin{aligned} \left\langle \frac{\cos kx}{\sqrt{\pi}}, \frac{\sin kx}{\sqrt{\pi}} \right\rangle &= \int_{-\pi}^{\pi} \frac{\cos kx}{\sqrt{\pi}} \cdot \frac{\sin kx}{\sqrt{\pi}} dx \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin kx \cos kx dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \sin 2kx dx \\ &= \frac{1}{2\pi} \left[-\frac{1}{2k} \cos 2kx \right]_{-\pi}^{\pi} = 0. \end{aligned}$$

Thus, every two different vectors in the list are orthogonal to each other. Hence, the list is an orthonormal list. ■

Problem 3

On $\mathcal{P}_2(\mathbb{R})$, consider the inner product given by

$$\langle p, q \rangle = \int_0^1 p(x) q(x) dx$$

Apply the Gram-Schmidt Procedure to the basis $1, x, x^2$ to produce an orthonormal basis of $\mathcal{P}_2(\mathbb{R})$.

✓ **Solution:** Let $\alpha_1 = 1$. Then

$$\|\alpha_1\| = \sqrt{\int_0^1 1 dx} = 1,$$

so the first vector of the orthonormal basis is $\beta_1 = 1$.

Now, let $\alpha_2 = x$ and compute:

$$\beta_2 = \frac{\alpha_2 - \langle \alpha_2, \beta_1 \rangle \beta_1}{\|\alpha_2 - \langle \alpha_2, \beta_1 \rangle \beta_1\|}, \quad \langle \alpha_2, \beta_1 \rangle = \int_0^1 x dx = \frac{1}{2}, \quad \alpha_2 - \langle \alpha_2, \beta_1 \rangle \beta_1 = x - \frac{1}{2},$$

$$\left\|x - \frac{1}{2}\right\| = \sqrt{\int_0^1 \left(x - \frac{1}{2}\right)^2 dx} = \sqrt{\int_0^1 \left(x^2 - x + \frac{1}{4}\right) dx} = \sqrt{\frac{1}{3} - \frac{1}{2} + \frac{1}{4}} = \frac{1}{2\sqrt{3}},$$

$$\Rightarrow \beta_2 = 2\sqrt{3}x - \sqrt{3}.$$

Let $\alpha_3 = x^2$. Then,

$$\beta_3 = \frac{\alpha_3 - \langle \alpha_3, \beta_1 \rangle \beta_1 - \langle \alpha_3, \beta_2 \rangle \beta_2}{\|\alpha_3 - \langle \alpha_3, \beta_1 \rangle \beta_1 - \langle \alpha_3, \beta_2 \rangle \beta_2\|}.$$

We do the orthogonalization first:

$$\langle \alpha_3, \beta_1 \rangle = \int_0^1 x^2 dx = \frac{1}{3}, \quad \langle \alpha_3, \beta_2 \rangle = \int_0^1 (2\sqrt{3}x^3 - \sqrt{3}x^2) dx = \frac{\sqrt{3}}{6},$$

$$\Rightarrow \alpha_3 - \langle \alpha_3, \beta_1 \rangle \beta_1 - \langle \alpha_3, \beta_2 \rangle \beta_2 = x^2 - x + \frac{1}{6}.$$

Now compute the norm for normalization:

$$\begin{aligned} \left\|x^2 - x + \frac{1}{6}\right\| &= \sqrt{\int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx} \\ &= \sqrt{\int_0^1 \left(x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36}\right) dx} \\ &= \sqrt{\frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36}} = \frac{1}{6\sqrt{5}}, \\ &\Rightarrow \beta_3 = 6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5}. \end{aligned}$$

Therefore, the orthonormal basis consists of:

$$1, \quad 2\sqrt{3}x - \sqrt{3}, \quad 6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5}.$$

Problem 4

For each of the following, use the Gram-Schmidt process find an orthonormal basis for $R(A)$:

$$1. A = \begin{bmatrix} -1 & 3 \\ 1 & 5 \end{bmatrix}$$

$$2. A = \begin{bmatrix} 2 & 5 \\ 1 & 10 \end{bmatrix}$$

where $R(A)$ is the linear space spanned by the columns of A .

✓ **Solution:**

$$1. \text{ To get started, we have } \left\| \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\| = \sqrt{2}, \text{ and hence } e_1 = \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Now the numerator in the expression for e_2 is

$$\begin{pmatrix} 3 \\ 5 \end{pmatrix} - \left\langle \begin{pmatrix} 3 \\ 5 \end{pmatrix}, e_1 \right\rangle e_1 = \begin{pmatrix} 3 \\ 5 \end{pmatrix} - \left(3 \cdot \frac{-1}{\sqrt{2}} + 5 \cdot \frac{1}{\sqrt{2}} \right) \cdot \begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix}.$$

We have $\left\| \begin{pmatrix} 4 \\ 4 \end{pmatrix} \right\| = 4\sqrt{2}$, and hence $e_2 = \frac{1}{4\sqrt{2}} \begin{pmatrix} 4 \\ 4 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$.

Thus

$$\begin{pmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

is an orthonormal list of length 2 in $R(A)$, and hence an orthonormal basis of $R(A)$.

2. To get started, we have $\left\| \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\| = \sqrt{5}$, and hence $e_1 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$.

Now the numerator in the expression for e_2 is

$$\begin{pmatrix} 5 \\ 10 \end{pmatrix} - \left\langle \begin{pmatrix} 5 \\ 10 \end{pmatrix}, e_1 \right\rangle e_1 = \begin{pmatrix} 5 \\ 10 \end{pmatrix} - \left(5 \cdot \frac{2}{\sqrt{5}} + 10 \cdot \frac{1}{\sqrt{5}} \right) \cdot \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \end{pmatrix}.$$

We have $\left\| \begin{pmatrix} -3 \\ 6 \end{pmatrix} \right\| = 3\sqrt{5}$, and hence $e_2 = \frac{1}{3\sqrt{5}} \begin{pmatrix} -3 \\ 6 \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$.

Thus

$$\begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}, \begin{pmatrix} -\frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{pmatrix}$$

is an orthonormal list of length 2 in $R(A)$, and hence an orthonormal basis of $R(A)$.

Problem 5

Given $\mathbf{x}_1 = \frac{1}{2} (1, 1, 1, -1)^T$ and $\mathbf{x}_2 = \frac{1}{6} (1, 1, 3, 5)^T$, verify that these vectors form an orthonormal set in $\mathbb{R}^4$. Extend this set to an orthonormal basis for $\mathbb{R}^4$ by finding an orthonormal basis for the null space of

$$\begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{bmatrix}$$

[Hint: First find a basis for the null space and then use the Gram-Schmidt process.]

✓ Solution:

We have $\|\mathbf{x}_1\| = \sqrt{\frac{1}{4} (1^2 + 1^2 + 1^2 + (-1)^2)} = 1$, $\|\mathbf{x}_2\| = \sqrt{\frac{1}{36} (1^2 + 1^2 + 3^2 + 5^2)} = 1$, and

$$\langle \mathbf{x}_1, \mathbf{x}_2 \rangle = \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{3}{6} + \frac{-1}{2} \cdot \frac{5}{6} = 0.$$

Thus, these vectors form an orthonormal set in $\mathbb{R}^4$.

We have

$$\begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{bmatrix}.$$

Thus, we have a basis of the null space:

$$\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 0 \\ -3 \\ 1 \end{pmatrix}.$$

From the Fundamental Theorem of Linear Algebra, the null space is orthogonal to $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$. So we only need to apply the Gram-Schmidt Procedure to the basis of the null space.

To get started, we have $\left\| \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\| = \sqrt{2}$, and hence $\mathbf{x}_3 = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{pmatrix}$.

Now the numerator in the expression for $\mathbf{x}_4$ is

$$\begin{pmatrix} 4 \\ 0 \\ -3 \\ 1 \end{pmatrix} - \left\langle \begin{pmatrix} 4 \\ 0 \\ -3 \\ 1 \end{pmatrix}, \mathbf{x}_3 \right\rangle \mathbf{x}_3 = \begin{pmatrix} 4 \\ 0 \\ -3 \\ 1 \end{pmatrix} - (-2\sqrt{2}) \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -3 \\ 1 \end{pmatrix}.$$

We have $\left\| \begin{pmatrix} 2 \\ 2 \\ -3 \\ 1 \end{pmatrix} \right\| = 3\sqrt{2}$, and hence $\mathbf{x}_4 = \left(\frac{\sqrt{2}}{3}, \frac{\sqrt{2}}{3}, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6} \right)^T$.

Thus $\mathbf{x}_3, \mathbf{x}_4$ is an orthonormal basis of the null space.

Hence, $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4$, *i.e.*,

$$\frac{1}{2} (1, 1, 1, -1)^T, \frac{1}{6} (1, 1, 3, 5)^T, \left(\frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, 0 \right)^T, \left(\frac{\sqrt{2}}{3}, \frac{\sqrt{2}}{3}, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{6} \right)^T$$

is an orthonormal basis of $\mathbb{R}^4$.

Problem 6

Find a polynomial $q \in \mathcal{P}_2(\mathbb{R})$ such that

$$p\left(\frac{1}{2}\right) = \int_0^1 p(x) q(x) dx$$

for every $p \in \mathcal{P}_2(\mathbb{R})$.

✓ **Solution1:** *It is recommended that this solution be used.*

Let $\varphi(p) = p\left(\frac{1}{2}\right)$. From the proof of the Riesz representation theorem, using the orthonormal basis from Problem 3 (denote as e_1, e_2, e_3), we have

$$\begin{aligned}
 q(x) &= \overline{\varphi(e_1)}e_1 + \overline{\varphi(e_2)}e_2 + \overline{\varphi(e_3)}e_3 \\
 &= e_1\left(\frac{1}{2}\right)e_1 + e_2\left(\frac{1}{2}\right)e_2 + e_3\left(\frac{1}{2}\right)e_3 \quad (\text{the conjugate is not needed on } \mathbb{R}) \\
 &= 1 \cdot 1 + \left(2\sqrt{3} \cdot \frac{1}{2} - \sqrt{3}\right) \cdot (2\sqrt{3}x - \sqrt{3}) \\
 &\quad + \left(6\sqrt{5} \cdot \left(\frac{1}{2}\right)^2 - 6\sqrt{5} \cdot \frac{1}{2} + \sqrt{5}\right) (6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5}) \\
 &= 1 + 0 - \frac{\sqrt{5}}{2} (6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5}) \\
 &= -15x^2 + 15x - \frac{3}{2}.
 \end{aligned}$$

✓ **Solution2:**

Suppose $q(x) = a_0 + a_1x + a_2x^2$ (a_0, a_1, a_2 are constants), and $p(x) = k_0 + k_1x + k_2x^2$, $k_0, k_1, k_2 \in \mathbb{R}$. Thus, $p\left(\frac{1}{2}\right) = k_0 + \frac{1}{2}k_1 + \frac{1}{4}k_2$.

We have

$$\begin{aligned}
 \int_0^1 p(x)q(x)dx &= \int_0^1 [(k_0 + k_1x + k_2x^2) \cdot (a_0 + a_1x + a_2x^2)]dx \\
 &= \int_0^1 [a_0k_0 + (a_1k_0 + a_0k_1)x + (a_2k_0 + a_1k_1 + a_0k_2)x^2 + (a_2k_1 + a_1k_2)x^3 + a_2k_2x^4]dx \\
 &= \left[a_0k_0x + \frac{1}{2}(a_1k_0 + a_0k_1)x^2 + \frac{1}{3}(a_2k_0 + a_1k_1 + a_0k_2)x^3 + \frac{1}{4}(a_2k_1 + a_1k_2)x^4 + \frac{1}{5}a_2k_2x^5 \right]_0^1 \\
 &= \left(a_0 + \frac{1}{2}a_1 + \frac{1}{3}a_2 \right) k_0 + \left(\frac{1}{2}a_0 + \frac{1}{3}a_1 + \frac{1}{4}a_2 \right) k_1 + \left(\frac{1}{3}a_0 + \frac{1}{4}a_1 + \frac{1}{5}a_2 \right) k_2.
 \end{aligned}$$

Thus for any $k_0, k_1, k_2 \in \mathbb{R}$,

$$\left(a_0 + \frac{1}{2}a_1 + \frac{1}{3}a_2 \right) k_0 + \left(\frac{1}{2}a_0 + \frac{1}{3}a_1 + \frac{1}{4}a_2 \right) k_1 + \left(\frac{1}{3}a_0 + \frac{1}{4}a_1 + \frac{1}{5}a_2 \right) k_2 = k_0 + \frac{1}{2}k_1 + \frac{1}{4}k_2,$$

Compare the corresponding coefficients and we have

$$\begin{cases} a_0 + \frac{1}{2}a_1 + \frac{1}{3}a_2 = 1 \\ \frac{1}{2}a_0 + \frac{1}{3}a_1 + \frac{1}{4}a_2 = \frac{1}{2} \\ \frac{1}{3}a_0 + \frac{1}{4}a_1 + \frac{1}{5}a_2 = \frac{1}{4} \end{cases} \implies \begin{cases} a_0 = -\frac{3}{2} \\ a_1 = 15 \\ a_2 = -15 \end{cases}$$

Hence, $q(x) = -\frac{3}{2} + 15x - 15x^2$.

Problem 7

Find a polynomial $q \in \mathcal{P}_2(\mathbb{R})$ such that

$$\int_0^1 p(x) (\cos \pi x) dx = \int_0^1 p(x) q(x) dx$$

for every $p \in \mathcal{P}_2(\mathbb{R})$.

✓ **Solution1:** *It is recommended that this solution be used.*

Let $\varphi(p) = \int_0^1 p(x)(\cos \pi x)dx$. Using the orthonormal basis from Problem 3, we have

$$\begin{aligned} q(x) = & \underbrace{\left(\int_0^1 1(\cos \pi x)dx \right)}_{\varphi(1)} \cdot 1 + \underbrace{\left(\int_0^1 (2\sqrt{3}x - \sqrt{3})(\cos \pi x)dx \right)}_{\varphi(2\sqrt{3}x - \sqrt{3})} \cdot (2\sqrt{3}x - \sqrt{3}) \\ & + \underbrace{\left(\int_0^1 (6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5})(\cos \pi x)dx \right)}_{\varphi(6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5})} \cdot (6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5}), \end{aligned}$$

where $\int_0^1 1(\cos \pi x)dx = 0$,

$$\int_0^1 x \cos \pi x dx = \frac{1}{\pi} \int_0^1 x d(\sin \pi x) = \frac{1}{\pi} \left(x \sin \pi x \Big|_0^1 - \int_0^1 \sin \pi x dx \right) = -\frac{2}{\pi^2},$$

and

$$\begin{aligned} \int_0^1 x^2 \cos \pi x dx &= \frac{1}{\pi} \int_0^1 x^2 d(\sin \pi x) = \frac{1}{\pi} \left(x^2 \sin \pi x \Big|_0^1 - 2 \int_0^1 x \sin \pi x dx \right) \\ &= \frac{2}{\pi^2} \left(x \cos \pi x \Big|_0^1 - \int_0^1 \cos \pi x dx \right) = -\frac{2}{\pi^2}. \end{aligned}$$

Thus, we have

$$q(x) = 0 \cdot 1 - \frac{4\sqrt{3}}{\pi^2} (2\sqrt{3}x - \sqrt{3}) + 0 \cdot (6\sqrt{5}x^2 - 6\sqrt{5}x + \sqrt{5}) = -\frac{12}{\pi^2} (2x - 1).$$

✓ **Solution2:**

Suppose $q(x) = a_0 + a_1x + a_2x^2$ (a_0, a_1, a_2 are constants), and $p(x) = k_0 + k_1x + k_2x^2, k_0, k_1, k_2 \in \mathbb{R}$.

Thus, we have

$$\begin{aligned} \int_0^1 p(x) (\cos \pi x) dx &= \int_0^1 (k_0 + k_1x + k_2x^2) (\cos \pi x) dx \\ &= k_0 \int_0^1 \cos \pi x dx + k_1 \int_0^1 x \cos \pi x dx + k_2 \int_0^1 x^2 \cos \pi x dx \\ &= 0 \cdot k_0 + \left(-\frac{2}{\pi^2} \right) \cdot k_1 + \left(-\frac{2}{\pi^2} \right) \cdot k_2. \end{aligned}$$

From **Solution2** of Problem 6, for any $k_0, k_1, k_2 \in \mathbb{R}$, we have

$$\int_0^1 p(x) q(x) dx = \left(a_0 + \frac{1}{2}a_1 + \frac{1}{3}a_2 \right) k_0 + \left(\frac{1}{2}a_0 + \frac{1}{3}a_1 + \frac{1}{4}a_2 \right) k_1 + \left(\frac{1}{3}a_0 + \frac{1}{4}a_1 + \frac{1}{5}a_2 \right) k_2.$$

Compare the corresponding coefficients and we have

$$\begin{cases} a_0 + \frac{1}{2}a_1 + \frac{1}{3}a_2 = 0 \\ \frac{1}{2}a_0 + \frac{1}{3}a_1 + \frac{1}{4}a_2 = -\frac{2}{\pi^2} \\ \frac{1}{3}a_0 + \frac{1}{4}a_1 + \frac{1}{5}a_2 = -\frac{2}{\pi^2} \end{cases} \Rightarrow \begin{cases} a_0 = \frac{12}{\pi^2} \\ a_1 = -\frac{24}{\pi^2} \\ a_2 = 0 \end{cases}$$

Hence, $q(x) = \frac{12}{\pi^2} - \frac{24}{\pi^2}x$.

Problem 8

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 12 \\ 6 \\ 18 \end{bmatrix}$$

- (a) Use the Gram-Schmidt process to find an orthonormal basis for the column space of A .
- (b) Factor A into a product QR , where Q has an orthonormal set of column vectors and R is upper triangular.
- (c) Solve the least squares problem $A\mathbf{x} = \mathbf{b}$.

✓ **Solution:** (a) Apply the Gram-Schmidt process to the columns of A , and record the coefficients that will form R . Let

$$\mathbf{a}_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Compute the norm of $\mathbf{a}_1$ (this will be r_{11}):

$$r_{11} = \|\mathbf{a}_1\| = \sqrt{2^2 + 1^2 + 2^2} = 3,$$

and obtain the first orthonormal vector

$$\mathbf{e}_1 = \frac{\mathbf{a}_1}{r_{11}} = \left(\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right)^T.$$

Calculate the projection coefficient of $\mathbf{a}_2$ onto $\mathbf{e}_1$:

$$r_{12} = \langle \mathbf{a}_2, \mathbf{e}_1 \rangle = 1 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{5}{3}.$$

Subtract the projection to obtain a vector orthogonal to $\mathbf{e}_1$:

$$\mathbf{u}_2 = \mathbf{a}_2 - r_{12}\mathbf{e}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \frac{5}{3} \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix} = \left(-\frac{1}{9}, \frac{4}{9}, -\frac{1}{9} \right)^T.$$

Find the norm of $\mathbf{u}_2$ (this will be r_{22}):

$$r_{22} = \|\mathbf{u}_2\| = \sqrt{\frac{1}{81} + \frac{16}{81} + \frac{1}{81}} = \frac{\sqrt{2}}{3},$$

and the second orthonormal vector

$$\mathbf{e}_2 = \frac{\mathbf{u}_2}{r_{22}} = \left(-\frac{\sqrt{2}}{6}, \frac{2\sqrt{2}}{3}, -\frac{\sqrt{2}}{6} \right)^T.$$

Thus $\{\mathbf{e}_1, \mathbf{e}_2\}$ is an orthonormal basis for the column space of A .

(b) From the construction we directly have

$$\mathbf{a}_1 = r_{11}\mathbf{e}_1, \quad \mathbf{a}_2 = r_{12}\mathbf{e}_1 + r_{22}\mathbf{e}_2,$$

which in matrix form is $A = QR$ with

$$Q = \begin{bmatrix} \frac{2}{3} & -\frac{\sqrt{2}}{6} \\ \frac{1}{3} & \frac{2\sqrt{2}}{3} \\ \frac{2}{3} & -\frac{\sqrt{2}}{6} \end{bmatrix}, \quad R = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix} = \begin{bmatrix} 3 & \frac{5}{3} \\ 0 & \frac{\sqrt{2}}{3} \end{bmatrix}.$$

Q has orthonormal columns and R is upper triangular; the entries of R were obtained during the Gram-Schmidt process.

(c) The least squares problem can be solved in two ways.

Method 1 (Normal equations).

$$A^T A = \begin{bmatrix} 9 & 5 \\ 5 & 3 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 66 \\ 36 \end{bmatrix},$$

and solving $(A^T A)\hat{\mathbf{x}} = A^T \mathbf{b}$ gives

$$\hat{\mathbf{x}} = \begin{bmatrix} 9 & 5 \\ 5 & 3 \end{bmatrix}^{-1} \begin{bmatrix} 66 \\ 36 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & -\frac{5}{2} \\ -\frac{5}{2} & \frac{9}{2} \end{bmatrix} \begin{bmatrix} 66 \\ 36 \end{bmatrix} = \begin{bmatrix} 9 \\ -3 \end{bmatrix}.$$

Method 2 (QR decomposition). Because $Q^T Q = I$, the normal equations reduce to $R\hat{\mathbf{x}} = Q^T \mathbf{b}$. First compute

$$Q^T \mathbf{b} = \begin{bmatrix} \mathbf{e}_1 \cdot \mathbf{b} \\ \mathbf{e}_2 \cdot \mathbf{b} \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \cdot 12 + \frac{1}{3} \cdot 6 + \frac{2}{3} \cdot 18 \\ -\frac{\sqrt{2}}{6} \cdot 12 + \frac{2\sqrt{2}}{3} \cdot 6 - \frac{\sqrt{2}}{6} \cdot 18 \end{bmatrix} = \begin{bmatrix} 22 \\ -\sqrt{2} \end{bmatrix}.$$

Now solve the upper triangular system $R\hat{\mathbf{x}} = Q^T \mathbf{b}$:

$$\begin{bmatrix} 3 & \frac{5}{3} \\ 0 & \frac{\sqrt{2}}{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 22 \\ -\sqrt{2} \end{bmatrix}.$$

From the second row, $\frac{\sqrt{2}}{3}x_2 = -\sqrt{2} \implies x_2 = -3$. Substituting into the first row: $3x_1 + \frac{5}{3}(-3) = 22 \implies x_1 = 9$. Hence

$$\hat{\mathbf{x}} = \begin{bmatrix} 9 \\ -3 \end{bmatrix}.$$

Problem 9

Suppose $v_1, \dots, v_m \in V$. Prove that

$$\{v_1, \dots, v_m\}^\perp = (\text{span}(v_1, \dots, v_m))^\perp$$

✓ **Proof:** Let $u \in \{v_1, \dots, v_m\}^\perp$, then we have

$$u \perp v_i, \quad i = 1, \dots, m,$$

which implies that $\langle v_1, u \rangle = \dots = \langle v_m, u \rangle = 0$. Thus, for any $v \in \text{span}(v_1, \dots, v_m)$, which can be written as $v = a_1 v_1 + \dots + a_m v_m$, where $a_1, \dots, a_m \in \mathbb{F}$, we have

$$\begin{aligned} \langle v, u \rangle &= \langle a_1 v_1 + \dots + a_m v_m, u \rangle \\ &= a_1 \langle v_1, u \rangle + \dots + a_m \langle v_m, u \rangle \\ &= a_1 \cdot 0 + \dots + a_m \cdot 0 \\ &= 0, \end{aligned}$$

which implies that $u \in (\text{span}(v_1, \dots, v_m))^\perp$. Hence, $\{v_1, \dots, v_m\}^\perp \subseteq (\text{span}(v_1, \dots, v_m))^\perp$.

Conversely, if $u \in (\text{span}(v_1, \dots, v_m))^\perp$, since $v_i \in \text{span}(v_1, \dots, v_m)$, $i = 1, \dots, m$, we have

$$u \perp v_i, \quad i = 1, \dots, m.$$

Thus, $u \in \{v_1, \dots, v_m\}^\perp$. Hence, $(\text{span}(v_1, \dots, v_m))^\perp \subseteq \{v_1, \dots, v_m\}^\perp$.

Hence,

$$\{v_1, \dots, v_m\}^\perp = (\text{span}(v_1, \dots, v_m))^\perp. \quad \blacksquare$$

Problem 10

Suppose U is the subspace of $\mathbb{R}^4$ defined by

$$U = \text{span}((1, 2, 3, -4), (-5, 4, 3, 2)).$$

Find an orthonormal basis of U and an orthonormal basis of $U^\perp$.

✓ **Solution:** To get started, we have $\alpha_1 = (1, 2, 3, -4)$, $\alpha_2 = (-5, 4, 3, 2)$. Then we get

$$\beta_1 = \frac{1}{\|\alpha_1\|} \alpha_1 = \frac{1}{\sqrt{30}} \alpha_1 = \left(\frac{1}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{3}{\sqrt{30}}, -\frac{4}{\sqrt{30}} \right).$$

To get β_2 , we compute it step-by-step:

$$\begin{aligned} \langle \alpha_2, \beta_1 \rangle &= \frac{4}{\sqrt{30}}, \quad \langle \alpha_2, \beta_1 \rangle \beta_1 = \left(\frac{2}{15}, \frac{4}{15}, \frac{2}{5}, -\frac{8}{15} \right), \\ \alpha_2 - \langle \alpha_2, \beta_1 \rangle \beta_1 &= \left(-\frac{77}{15}, \frac{56}{15}, \frac{13}{5}, \frac{38}{15} \right), \\ \beta_2 &= \frac{\alpha_2 - \langle \alpha_2, \beta_1 \rangle \beta_1}{\|\alpha_2 - \langle \alpha_2, \beta_1 \rangle \beta_1\|} = \frac{15}{\sqrt{12030}} \left(-\frac{77}{15}, \frac{56}{15}, \frac{13}{5}, \frac{38}{15} \right). \end{aligned}$$

Now we get an orthonormal basis of U :

$$\beta_1 = \frac{1}{\sqrt{30}} (1, 2, 3, -4), \beta_2 = \frac{1}{\sqrt{12030}} (-77, 56, 39, 38).$$

To find an orthonormal basis of $U^\perp$, we let $A = \begin{pmatrix} 1 & 2 & 3 & -4 \\ -5 & 4 & 3 & 2 \end{pmatrix}$. Solve $Ax = 0$, and then we can get two linearly independent vectors that span the null space of A , which is also $U^\perp$. Such two vectors form a basis of $U^\perp$.

We have

$$\begin{pmatrix} 1 & 2 & 3 & -4 \\ -5 & 4 & 3 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & -4 \\ 0 & 14 & 18 & -18 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 & -4 \\ 0 & 1 & \frac{9}{7} & -\frac{9}{7} \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & \frac{3}{7} & -\frac{10}{7} \\ 0 & 1 & \frac{9}{7} & -\frac{9}{7} \end{pmatrix}$$

Thus, we have a basis of $U^\perp$:

$$x_1 = \left(-\frac{3}{7}, -\frac{9}{7}, 1, 0\right), x_2 = \left(\frac{10}{7}, \frac{9}{7}, 0, 1\right).$$

Then we apply the Gram-Schmidt process to these two vectors. The step-by-step computation is shown as follows.

$$\begin{aligned} \gamma_1 &= \frac{1}{\|x_1\|} x_1 = \frac{7}{\sqrt{139}} \left(-\frac{3}{7}, -\frac{9}{7}, 1, 0\right) \\ \langle x_2, \gamma_1 \rangle &= -\frac{7}{\sqrt{139}} \cdot \frac{111}{49} = -\frac{111}{7\sqrt{139}} \\ \langle x_2, \gamma_1 \rangle \gamma_1 &= -\frac{111}{7\sqrt{139}} \frac{7}{\sqrt{139}} \left(-\frac{3}{7}, -\frac{9}{7}, 1, 0\right) = -\frac{111}{139} \left(-\frac{3}{7}, -\frac{9}{7}, 1, 0\right) = \left(\frac{333}{973}, \frac{999}{973}, -\frac{111}{139}, 0\right) \\ x_2 - \langle x_2, \gamma_1 \rangle \gamma_1 &= \left(\frac{151}{139}, \frac{36}{139}, \frac{111}{139}, 1\right) \\ \gamma_2 &= \frac{x_2 - \langle x_2, \gamma_1 \rangle \gamma_1}{\|x_2 - \langle x_2, \gamma_1 \rangle \gamma_1\|} = \frac{139}{\sqrt{55739}} \left(\frac{151}{139}, \frac{36}{139}, \frac{111}{139}, 1\right) \end{aligned}$$

Now we get an orthonormal basis of $U^\perp$:

$$\gamma_1 = \frac{1}{\sqrt{139}} (-3, -9, 7, 0), \gamma_2 = \frac{1}{\sqrt{55739}} (151, 36, 111, 139).$$

! Note: We can also add the standard basis of $\mathbb{R}^4$ to the basis of U and apply Gram-Schmidt process to them, and we get $\frac{1}{\sqrt{76190}}(190, 117, 60, 151), \frac{1}{\sqrt{190}}(0, 9, -10, -3)$ as an orthonormal basis of $U^\perp$.

Problem 11

Let U be an m -dimensional subspace of $\mathbb{R}^n$ and let V be a k -dimensional subspace of U , where $0 < k < m$.

(a) Show that any orthonormal basis

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$$

for V can be expanded to form an orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_m\}$ for U .

(b) Show that if $W = \text{Span}\{\mathbf{v}_{k+1}, \dots, \mathbf{v}_m\}$, then $U = V \oplus W$.

✓ **Proof:** (a) Since $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is an orthonormal list in U , it is linearly independent. Thus we can extend it to a basis of U :

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{u}_{k+1}, \dots, \mathbf{u}_m\}$$

And then apply the Gram-Schmidt procedure to it and get an orthonormal basis of U :

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_m\}.$$

Here the Gram-Schmidt procedure leaves the first m vectors unchanged because they are already orthonormal. Gram-Schmidt does not change the span of the list of vectors, thus we find the desired basis.

(b) There are at least two different proofs.

✓ **Proof1:** From the Gram-Schmidt procedure, it's seen that

$$\mathbf{v}_i \perp \mathbf{v}_j, \quad i = k+1, \dots, m, \quad j = 1, \dots, k,$$

which implies that

$$\mathbf{v}_i \in \{\mathbf{v}_1, \dots, \mathbf{v}_k\}^\perp, \quad i = k+1, \dots, m.$$

From Problem 9, we have

$$\{\mathbf{v}_1, \dots, \mathbf{v}_k\}^\perp = (\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k))^\perp,$$

Thus, $W \subseteq V^\perp$. Since $\{\mathbf{v}_{k+1}, \dots, \mathbf{v}_m\}$ is a basis of W , we have $\dim W = m-k$. Since $\dim U = m$, $\dim V = k$, and $U = V \oplus V^\perp$, we have $\dim V^\perp = m-k = \dim W$. Thus, $W = V^\perp$. Hence, $U = V \oplus W$.

✓ **Proof2:** First we show that $U = V + W$. From (a), we have $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_m$ is an orthonormal basis of U , then every vector $\mathbf{u} \in U$ can be written as

$$\mathbf{u} = \underbrace{a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k}_{\mathbf{v}} + \underbrace{a_{k+1} \mathbf{v}_{k+1} + \dots + a_m \mathbf{v}_m}_{\mathbf{w}},$$

where $\mathbf{v} \in V$ and $\mathbf{w} \in W$. Thus, $U = V + W$.

To show that $U = V \oplus W$, we now need only show that $V \cap W = \{\mathbf{0}\}$. Suppose $\mathbf{u} \in V \cap W$. Then we have $\mathbf{u} = a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k = b_1 \mathbf{v}_{k+1} + \dots + b_{m-k} \mathbf{v}_m$, which implies that

$$a_1 \mathbf{v}_1 + \dots + a_k \mathbf{v}_k - (b_1 \mathbf{v}_{k+1} + \dots + b_{m-k} \mathbf{v}_m) = \mathbf{0}.$$

Since $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_m$ are linearly independent, the equation above holds if and only if $a_1 = \dots = a_k = b_1 = \dots = b_{m-k} = 0$, which implies $\mathbf{u} = \mathbf{0}$. Thus, $V \cap W = \{\mathbf{0}\}$. Hence, $U = V \oplus W$. ■